

2. Hostname

Function: `gethostname()`

```
</> C
char hostname[1024];
gethostname(hostname, sizeof(hostname));
```

Logic:

- `gethostname()` asks the OS for the **name of the machine**.
- Example:

```
Saanvis-MacBook-Air
```

You need:

```
</> C
#include <unistd.h>
```

how to get the current working directory, hostname and user name just give me the functions and explain the logic

If you're implementing your **C shell**, these are the three things you need:

1. Current working directory

Function: `getcwd()`

```
</> C
char cwd[1024];
getcwd(cwd, sizeof(cwd));
```

Logic:

- `getcwd()` asks the OS: "What directory is this process currently in?"
- It puts the path into your character array.
- Example result:

```
/home/saanvi/cshell
```

You need:

```
</> C
#include <unistd.h>
```

3. Username

For the username, use `getpwuid()`:

```
</> C
struct passwd *pw = getpwuid(getuid());
char *username = pw->pw_name;
```

Logic:

Think of it as two steps:

```
getuid()
↓
gets your user ID (UID)
↓
getpwuid()
↓
looks up information about that UID
↓
pw->pw_name
↓
username
```

For example:

```
UID = 501
↓
getpwuid(501)
↓
struct passwd
↓
pw_name = "saanvi"
```

Ask anything

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what does getcwd return, just give the man bro

`getcwd()` returns:

A pointer to the buffer containing the current working directory on success, and `NULL` on failure.

Man-page signature:

```
</> C
char *getcwd(char *buf, size_t size);
```

So:

```
</> C
char cwd[1024];

if (getcwd(cwd, sizeof(cwd)) != NULL) {
    printf("%s\n", cwd);
}
```

The return value is basically the same `buf` pointer you passed in.

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getuid?

```
</> C
uid_t getuid(void);
```

Returns the **real user ID (UID)** of the calling process.

- No arguments.
- Returns a `uid_t`.
- Example: `getuid()` → `501` on macOS, for example.

To get the username from it:

```
</> C
getpwuid(getuid())
```

`getuid()` itself does not return the username.

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now how do I get the relative path, how to format it

If you mean **display the current working directory relative to the user's home directory**, the logic is:

```
/home/saanvi/cshell
/home/saanvi
  ↓
~/cshell
```

Get the home directory using:

```
<> C
getpwuid(getuid())->pw_dir
```

Then compare it with the current working directory from `getcwd()`.

Formatting

If:

```
cwd = /home/saanvi/cshell
home = /home/saanvi
```

you want:

```
~/cshell
```

If `cwd == home`, just print:

```
~
```

just tell the return type of these and function signature

- `getcwd`

```
</> C
char *getcwd(char *buf, size_t size);
```

- `getuid`

```
</> C
uid_t getuid(void);
```

- `getpwuid`

```
</> C
struct passwd *getpwuid(uid_t uid);
```

- `gethostname`

```
</> C
int gethostname(char *name, size_t namelen);
```

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how to store uid in string and home dir in string

`getuid()` returns a `uid_t`, not a string. If you actually need the **UID as a string**:

```
</> C
char uid_str[20];
snprintf(uid_str, sizeof(uid_str), "%d", getuid());
```

For the **home directory**:

```
</> C
struct passwd *pw = getpwuid(getuid());
char *home = pw->pw_dir;
```

Here, `home` already points to a string. You don't need to copy it unless you specifically want your own buffer.

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now how do i get relaive path

If you mean **relative to the home directory**, you already have:

```
</> C
char *home = getpwuid(getuid())->pw_dir;
char cwd[PATH_MAX];
getcwd(cwd, sizeof(cwd));
```

Then:

```
</> C
char *relative;
if (strcmp(cwd, home) == 0) {
    relative = "~";
} else if (strncmp(cwd, home, strlen(home)) == 0) {
    relative = cwd + strlen(home);
}
```

But note: if `cwd = "/Users/saanvi/projects"`, `relative` becomes:

```
/projects
```

So for the shell prompt you would format it as:

```
~ + relative
```

giving:



If the current working directory has the home directory as an ancestor (e.g. `/path/to/home/osn`), replace the home directory prefix with `~`. So the example becomes `~/osn`.

this is not happening
how to fix this

Share ...

You need to do a **prefix replacement**: if the current directory starts with `$HOME`, replace that prefix with `~`.

In C, the logic is:

```
<> C
char cwd[PATH_MAX];
char *home = getenv("HOME");

getcwd(cwd, sizeof(cwd));

if (strncmp(cwd, home, strlen(home)) == 0) {
    // cwd is inside HOME
}
```

But there is an important edge case: `/home/user2` must not match `/home/user`.

So check that the character immediately after the home prefix is either `/` or `'\0'`:

```
<> C
if (strncmp(cwd, home, strlen(home)) == 0 &&
```

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```
</> C
}
```



Sha

But there is an important edge case: `/home/user2` must **not** match `/home/user` .

So check that the character immediately after the home prefix is either `/` or `'\0'` :

```
</> C
```



```
if (strcmp(cwd, home, strlen(home)) == 0 &&
    (cwd[strlen(home)] == '/' || cwd[strlen(home)] == '\0')) {
    // replace home prefix with "~"
}
```

For example:

```
HOME = /Users/saanvi

cwd = /Users/saanvi/osn
    ↓
result = ~/osn
```



How to actually construct the string

You don't need to modify `cwd` in place. Create another buffer:

```
</> C
```



```
char display_path[PATH_MAX];

if (...) {
    snprintf(display_path, sizeof(display_path), "~%s",
             cwd + strlen(home));
} else {
```

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Think



```
</> C

char display_path[PATH_MAX];

if (...) {
    snprintf(display_path, sizeof(display_path), "%s",
              cwd + strlen(home));
} else {
    snprintf(display_path, sizeof(display_path), "%s", cwd);
}
```

The key part is:

```
</> C

cwd + strlen(home)
```

If:

```
cwd = /Users/saanvi/osn
home = /Users/saanvi
```

then:

```
cwd + strlen(home)
  ↓
/osn
```

So:

```
</> C

"^" + "/osn"
```

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The key part is:

```
</> C  
cwd + strlen(home)
```

If:

```
cwd = /Users/saanvi/osn  
home = /Users/saanvi
```

then:

```
cwd + strlen(home)  
↓  
/osn
```

So:

```
</> C  
"~" + "/osn"
```

becomes:

```
~/osn
```

Also: don't use `getuid()` to obtain the home directory. `getuid()` gives you the user's numeric UID. For the home path, use `getenv("HOME")` or look it up with `getpwuid()`.

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```
char *home = getenv("HOME");
```

Yes. This line:

```
</> C
char *home = getenv("HOME");
```

means:

- `getenv()` looks for an **environment variable** named `"HOME"`.
- It returns a `char *` pointing to the value of that variable.
- On your Mac, for example, it might return:

```
/Users/saanvi
```

So after:

```
</> C
char *home = getenv("HOME");
```

you can think of:

```
home
↓
"/Users/saanvi"
```

Function signature

you can think of:

```
home
↓
"/Users/saanvi"
```

Function signature

```
</> C
char *getenv(const char *name);
```

If the variable doesn't exist, it returns:

```
</> C
NULL
```

So it's good practice to check:

```
</> C
char *home = getenv("HOME");

if (home == NULL) {
    // HOME doesn't exist
}
```

For your shell prompt, you'll then compare `home` with the beginning of the current working directory obtained from `getcwd()`.

any other function which does this

Yes. If you mean another way to get the user's home directory, there are two common approaches.

1. `getpwuid()` — more robust

```
</> C
#include <pwd.h>
#include <unistd.h>

struct passwd *pw = getpwuid(getuid());

char *home = pw->pw_dir;
```

Here:

```
</> C
getuid()
```

→ gets the user's UID.

```
</> C
getpwuid(uid)
```

→ looks up that user's account information.

```
</> C
```

```
</> C
getpwuid(uid)
```

→ looks up that user's account information.

```
</> C
pw->pw_dir
```

→ gives the home directory.

So you get something like:

```
/Users/saanvi
```

2. `getenv("HOME")` — simplest

```
</> C
char *home = getenv("HOME");
```

For your **c-shell project**, I'd use `getenv("HOME")` for now. It's much simpler and perfectly suitable for implementing the `~` prompt behavior.

If you specifically want to practice Unix system calls / POSIX APIs, then `getpwuid(getuid())` is worth understanding.

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explain it all again how do file descriptors and pipe work

Yes. Let's restart from zero and connect **file descriptors** → `pipe()` → `fork()` → **shell pipe** |. The most important thing is to understand what is in **user space** versus what the **kernel** maintains.

1. What is a file descriptor?

A **file descriptor (FD)** is just an integer that a process uses to refer to an open I/O resource.

For example:

```
</> C
int fd = open("hello.txt", O_RDONLY);
```

Suppose the OS returns:

```
fd = 3
```

Your program now has the number `3`.

But **3 is not the file**.

The OS maintains a table for your process that is conceptually like:

```
PROCESS
File Descriptor Table
0 | —————> terminal
```

But 3 is not the file.

The OS maintains a table for your process that is conceptually like:

PROCESS
File Descriptor Table

0	→	terminal
1	→	terminal
2	→	terminal
3	→	hello.txt

So when you do:

```
<> C  
read(3, buffer, 100);
```

the kernel sees:

"This process is asking me to read from FD 3. FD 3 refers to `hello.txt`."

It then performs the actual read.

Therefore:

```
3  
↓  
FD table entry  
↓  
kernel's representation of the open resource  
↓  
hello.txt
```

hello.txt

Share

The FD is basically a **handle/index** that the process uses to access something managed by the kernel.

2. What are 0, 1 and 2?

Every normal process starts with these three:

```
0 → stdin
1 → stdout
2 → stderr
```



For a terminal:

```
      TERMINAL
     /   |   \
    /    |    \
   stdin stdout stderr
     0     1     2
```



So:

```
<> C
printf("hello");
```



ultimately writes to **FD 1**, stdout.

And:

```
<> C
```



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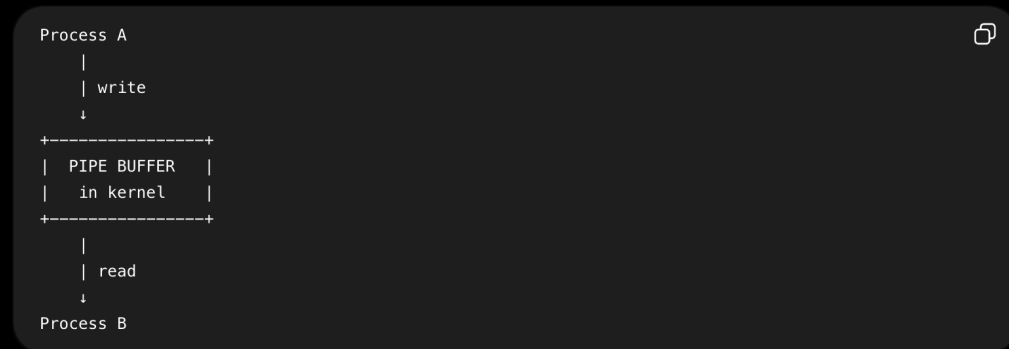
```
scanf(...);
```

reads from FD 0, stdin.

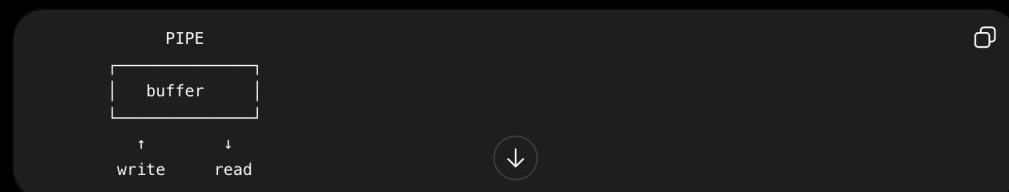
3. Now what is a pipe?

A pipe is a kernel-managed buffer that allows data to flow from one process to another.

Imagine:



The pipe has two ends:



write read

Share

One side writes data.

The other side reads data.

4. Creating a pipe

In C:

```
</> C
int fd[2];
pipe(fd);
```

This is slightly unusual because you're giving `pipe()` an array.

Before:

```
fd[0] = ?
fd[1] = ?
```

You call:

```
</> C
pipe(fd);
```

The kernel creates a pipe and gives you **two file descriptors**.

For example:

```
fd[1] = ?
```

Share

You call:

```
</> C
pipe(fd);
```

The kernel creates a pipe and gives you **two file descriptors**.

For example:

```
fd[0] = 3
fd[1] = 4
```

The convention is:

```
fd[0] → read end
fd[1] → write end
```

So now your process's FD table conceptually looks like:

```
PROCESS
FD TABLE
0  → terminal
1  → terminal
2  → terminal
3  → pipe READ end
4  → pipe WRITE end
```

And somewhere in kernel space:

5. Why does `pipe(fd)` modify `fd`?

Because `fd` is being used as an output parameter.

You allocate space:

```
</> C
int fd[2];
```

Then tell the kernel:

```
</> C
pipe(fd);
```

Meaning, conceptually:

"Create a pipe and put the two file descriptor numbers into this array."

So after the call:

```
fd[0] = read descriptor
fd[1] = write descriptor
```

For example:

```
fd[0] = 3
fd[1] = 4
```

what does clearerr do

The `clearerr()` function in C is a standard library function (defined in `<stdio.h>`) used to reset the end-of-file (EOF) and error flags on a file stream. [Programiz](#)

Syntax

```
C
void clearerr(FILE *stream);
```

Why does it exist?

When you read from a stream (like `stdin` or a file) in C, the system keeps internal status flags for that stream:

1. **The EOF Indicator:** Turns on automatically once your program hits the end of the file or stream. [GeeksforGeeks](#)
2. **The Error Indicator:** Turns on if a read/write operation fails. [Programiz](#)

The Catch: Once these flags are triggered, they stay set. If a stream hits EOF (like when you pressed `Ctrl+D` in your shell experiment), that stream is permanently marked as "dead/ended." If you try to read from it again later without clearing it, C will instantly assume you

"How do I set up a C project with separate `src/` and `include/` directories and a Makefile?"

Here is a clean, standard way to set up a C project with separate `src/` (implementation files) and `include/` (header files) directories, along with a robust Makefile.

1. Directory Structure

Organize your project directory like this:

text



```
my-c-project/
├── include/
│   ├── shell.h           # Central header (optional)
│   ├── prompt.h
│   └── utils.h
├── src/
│   ├── main.c
│   ├── prompt.c
│   └── utils.c
├── obj/                  # (Optional/Auto-generated) Object files .o
└── Makefile
```

2. Header and Source File Conventions

- In `include/prompt.h`: Use header guards and declare function prototypes and types.

2. Header and Source File Conventions

- In `include/prompt.h`: Use header guards and declare function prototypes and types.

```
C
#include <stdio.h>

#ifndef PROMPT_H
#define PROMPT_H

void display_prompt(void);

#endif /* PROMPT_H */
```

- In `src/prompt.c`: Include your header using quotes:

```
C
#include "prompt.h"
#include <stdio.h>

void display_prompt(void) {
    printf("<shell> ");
}
```

- In `src/main.c`:

```
C
#include "prompt.h"
```

3. Writing the Makefile

The key to separating headers and sources is the compiler flag `-Iinclude` (or `-I$(INC_DIR)`). This tells `gcc` / `clang` to search the `include/` directory for any `#include "header.h"` directives.

Here are two options depending on your preference:

Option A: Direct Build (Simple, like your current `c-shell/Makefile`)

Compiles all `.c` files directly into the executable in one step:

```
makefile

CC = gcc
CFLAGS = -Wall -Wextra -Werror -std=c11 -Iinclude

SRC_DIR = src
INC_DIR = include
TARGET = shell.out

SRCS = $(wildcard $(SRC_DIR)/*.c)

all: $(TARGET)

$(TARGET): $(SRCS)
    $(CC) $(CFLAGS) -o $(TARGET) $(SRCS)

clean:
    rm -f $(TARGET)

.PHONY: all clean
```

What compiler flags should I use for a C project that needs POSIX functions on macOS? I keep getting implicit declaration warnings



To fix implicit declaration warnings and successfully use POSIX functions on macOS, you need to tell the compiler which standards and feature test macros to adhere to. macOS (and BSD-based systems in general) hide POSIX and modern C library functions by default unless explicitly requested.

Here are the essential compiler flags you should use:

1. The Essential Flags

Add these flags to your compilation command (e.g., in your `CFLAGS` or Makefile):

- `-std=c99` (or `-std=c11`, `-std=c17`, etc.): Sets the C language standard.
- `-D_POSIX_C_SOURCE=200809L` : Tells the system headers to expose POSIX.1-2008 functions (such as `fork()`, `fileno()`, `sigaction`, etc.).
- `-D_DARWIN_C_SOURCE` : Enables Apple/Darwin-specific extensions alongside standard POSIX functions, which is often necessary on macOS for certain BSD/Darwin APIs.

2. Example Compilation Command

If you are compiling a file named `main.c`, your terminal command should look like this:

2. Example Compilation Command

If you are compiling a file named `main.c`, your terminal command should look like this:

Bash



```
clang -std=c11 -D_POSIX_C_SOURCE=200809L -D_DARWIN_C_SOURCE main.c -o my_program
```

If you are using a **Makefile**, define your flags like this:

Makefile



```
CFLAGS = -std=c11 -Wall -Wextra -D_POSIX_C_SOURCE=200809L -D_DARWIN_C_SOURCE
```

```
my_program: main.c
    $(CC) $(CFLAGS) main.c -o my_program
```

3. Crucial Rule: Header Inclusion Order

Even with the correct flags, you will still get implicit declaration warnings if you don't include the proper headers, or if you define feature macros *after* a header has already been included.

: [NOTE]

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Makefile



```
CFLAGS = -std=c11 -Wall -Wextra -D_POSIX_C_SOURCE=200809L -D_DARWIN_C_SOURCE

my_program: main.c
    $(CC) $(CFLAGS) main.c -o my_program
```

3. Crucial Rule: Header Inclusion Order

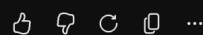
Even with the correct flags, you will still get implicit declaration warnings if you don't include the proper headers, or if you define feature macros *after* a header has already been included.

[!NOTE]

Always define `_POSIX_C_SOURCE` at the very top of your source file (before any `#include` directives), or pass it via the compiler command line (recommended, as shown above).

For example, to use functions like `popen()` or `fileno()`, ensure you include `<stdio.h>` and `<unistd.h>` after your compiler flags are applied.

Would you like help configuring this inside a specific build system like CMake or a Makefile?



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How do I split a string like `ls -la ; echo hello &` into tokens in C, handling operators like `;`, `&`, `>`, `>>`, `<`, and `|` as separate tokens?

To split a command string containing operators like `;`, `&`, `>`, `>>`, `<`, and `|` into separate tokens, standard functions like `strtok()` fall short because they drop delimiters or don't gracefully handle multi-character operators like `>>`.

Instead, writing a custom state-machine parser in C gives you precise control. It scans the string character-by-character, handles whitespace, groups regular words (or quotes, if needed), and treats special operators as standalone tokens.

Complete C Implementation

C



```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>

#define MAX_TOKENS 128
#define MAX_TOKEN_LEN 256

// Helper to check if a character is a whitespace
```

C



```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <stdbool.h>

#define MAX_TOKENS 128
#define MAX_TOKEN_LEN 256

// Helper to check if a character is a whitespace
bool is_whitespace(char c) {
    return c == ' ' || c == '\t' || c == '\n' || c == '\r';
}

// Helper to check if a character is a special operator starter
bool is_operator_char(char c) {
    return c == ';' || c == '&' || c == '|' || c == '<' || c == '>';
}

// Function to tokenize the input command string
int tokenize(char *input, char **tokens) {
    int token_count = 0;
    char *p = input;

    while (*p != '\0') {
        // 1. Skip whitespace
        if (is_whitespace(*p)) {
            p++;
            continue;
        }
    }
```

C



```
while (*p != '\0') {
    // 1. Skip whitespace
    if (is_whitespace(*p)) {
        p++;
        continue;
    }

    // Check if we have room for another token
    if (token_count >= MAX_TOKENS - 1) break;

    // 2. Handle Multi-character or Single-character Operators
    if (is_operator_char(*p)) {
        char *start = p;

        // Handle double-character operators like '>>' or '&&' or '||'
        if ((*p == '>' && *(p + 1) == '>') ||
            (*p == '&' && *(p + 1) == '&') ||
            (*p == '|' && *(p + 1) == '|')) {
            p += 2;
        } else {
            p += 1; // Single character operator like ';', '<', '|', '>'
        }

        // Allocate and copy the operator token
        int len = p - start;
        tokens[token_count] = malloc(len + 1);
        strncpy(tokens[token_count], start, len);
        tokens[token_count][len] = '\0';
        token_count++;
        continue;
    }
}
```

C



```
        continue;
    }

    // 3. Handle Regular Words/Arguments (e.g., ls, -la, hello)
    {
        char *start = p;
        // Read until we hit whitespace or an operator character
        while (*p != '\0' && !is_whitespace(*p) && !is_operator_char(*p)) {
            p++;
        }

        int len = p - start;
        tokens[token_count] = malloc(len + 1);
        strncpy(tokens[token_count], start, len);
        tokens[token_count][len] = '\0';
        token_count++;
    }
}

tokens[token_count] = NULL; // Null-terminate the array for execvp-style function
return token_count;
}

// Free allocated token memory
void free_tokens(char **tokens, int count) {
    for (int i = 0; i < count; i++) {
        free(tokens[i]);
    }
}

int main(void) {
    char command[] = "ls -la ; echo hello &";
    char *tokens[MAX_TOKENS];

    int count = tokenize(command, tokens);

    printf("Parsed %d tokens:\n", count);
    for (int i = 0; i < count; i++) {
        printf("Token %d: \"%s\"\n", i, tokens[i]);
    }

    free_tokens(tokens, count);
    return 0;
}
```

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How It Works

1. **Whitespace Skipping:** The pointer skips over spaces and tabs seamlessly.
2. **Operator Matching:** When it encounters an operator character (like `>`, `&`, or `|`), it checks ahead by 1 index to safely group double-character operators (`>>`) while leaving single operators (`>`) intact.
3. **Word Extraction:** Anything else is treated as a regular argument string and captured up until the next space or operator boundary.
4. **Memory Safety:** Each token is dynamically allocated using `malloc()`, ensuring you can safely store and pass them around (e.g., to process execution functions). Don't forget to call `free_tokens()` when done!

How do I handle edge cases in tokenization like multiple spaces, leading/trailing whitespace, and empty input?

The state-machine approach naturally handles most of these edge cases, but a few small adjustments make it completely robust against empty inputs, excessive whitespace, and buffer overflows.

Handling Each Edge Case

1. **Multiple Spaces & Leading/Trailing Whitespace:**
 - **Handled by:** The `while (is_whitespace(*p)) p++;` loop at the beginning of each iteration.
 - Leading spaces are skipped immediately. Consecutive spaces are consumed all at once, and trailing spaces are ignored when the pointer hits the null terminator (`\0`).
2. **Empty Input (`""` or `" "`):**
 - **Handled by:** If the string is empty or contains only whitespace, the loop never triggers token creation. `token_count` remains `0`, and the array is safely null-terminated.
3. **Buffer Limits (Safety Guard):**
 - **Handled by:** Ensuring that if the token limit is reached, the array is still properly null-terminated to prevent undefined behavior.

Refined Tokenizer Implementation

Here is the updated, hardened version of the `tokenize` function:

Implementing a hop command (like cd) in C using chdir(). It should handle hop, hop ~, hop -, hop .., and hop <path>. How do I track the previous directory for hop - (like cd - in bash)

To track the previous directory for a feature like `hop -`, your program needs to maintain a persistent memory buffer that acts similarly to the shell's `OLDPWD` environment variable. Since a custom shell or command loop runs as a continuous process, you can store this information in a dedicated variable outside of your command-processing function.

Here is how the logic works conceptually:

Core Tracking Mechanism

- **The Previous Directory Buffer:** Allocate a character array or string buffer in your main program scope to hold the absolute path of the directory you were in *before* your last successful directory change.
- **Capturing Before Changing:** Every time the user requests a directory change (other than `-`), your program should fetch the current working directory path *before* executing the change function.
- **Updating the State:** Once you successfully change directories, you update your previous directory buffer with that old path you just saved, so it is ready for the next time the user wants to jump back.

Handling `hop -` Specifically

1. When the user inputs `-`, your program checks if your previous directory buffer is currently empty (which happens if no directory change has occurred since the program started).
2. If it is empty, you inform the user that there is no previous directory to return to.
3. If it contains a valid path, you feed that stored path into your directory-changing function.
4. Crucially, before executing the change back, you must capture your *current* directory and swap it with your stored previous directory so that typing `-` repeatedly can toggle back and forth smoothly.

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Environment Integration

To make it behave just like a standard shell, you should also update the internal environment variables for the current working directory and the old working directory whenever a successful jump happens. This ensures any child processes or standard library calls are aware of the updated context.

Would you like to explore how to handle home directory expansions for `hop ~` using password database lookups?

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How does `chdir()` work? What happens if I pass it a path that doesn't exist?

How `chdir()` Works

The `chdir()` function is a fundamental system call in POSIX-compliant operating systems used to change the current working directory of the calling process.

- **Process-Specific Scope:** When a process invokes this function, the operating system updates the current working directory tracker for that specific running program. Any subsequent file operations (like opening, reading, or writing files using relative paths) will be evaluated relative to this new directory.
- **The Shell Caveat:** If you are building your own custom command-line shell, it is vital to remember that `chdir()` only changes the directory for the shell process itself. It cannot change the directory of the parent terminal window that launched your program, which is why shells must implement the directory change logic internally rather than as an external child process.

What Happens If a Path Doesn't Exist?

When you pass a path that does not exist to `chdir()`, the system call behaves predictably to protect your program's state:

- **Failure and Return Value:** The function fails and immediately returns an error indicator (specifically, a negative value, typically `-1`).
- **Preserving State:** The current working directory of your process remains completely unchanged. It does not crash or partially move; it simply stays wherever it was before the call was made.
- **Setting the Error Code:** Because it fails, it sets the global system error variable to indicate the exact cause. For a non-existent path, the error code corresponds to "No such file or directory." Other common errors you might encounter include permissions issues or trying to use a regular file as if it were a directory.

How do I print the owner and group name from uid and gid using `getpwuid` and `getgrgid`?

To print the owner name and group name from a user ID (`uid`) and group ID (`gid`), you use system database functions that query the operating system's user and group records.

Here is how the conceptual process works using `getpwuid` and `getgrgid` :

1. Retrieving the Owner Name (`getpwuid`)

- **The Function:** `getpwuid()` accepts a numeric user ID as its argument.
- **The Return Value:** It returns a pointer to a password structure containing information about that user.
- **Extracting the Name:** Inside that returned structure, look for the field designated to hold the user's login name string.
- **Safety Check:** If the user ID does not exist on the system (e.g., for orphaned files), the function returns a null pointer, so you should always check for that before trying to read the name string.

2. Retrieving the Group Name (`getgrgid`)

- **The Function:** `getgrgid()` accepts a numeric group ID as its argument.
- **The Return Value:** It returns a pointer to a group structure containing information about that specific group.
- **Extracting the Name:** Inside this structure, look for the field designated to hold the group name string.
- **Safety Check:** Just like with users, if the group ID is invalid or missing, it returns a null pointer, requiring a quick safety check.

How do I recursively traverse a directory tree in C using `opendir`, `readdir`, and `stat`?

To recursively traverse a directory tree in C using `opendir`, `readdir`, and `stat`, your program needs to combine directory iteration with a recursive function design.

Here is the conceptual step-by-step logic:

1. Opening the Directory

- Pass the path of the target directory to `opendir()`.
- Always check if the return value is valid (not null) to handle permission errors or non-existent paths safely before proceeding.

2. Iterating Through Entries

- Use a loop with `readdir()` to read the directory entries one by one until it returns a null pointer indicating the end of the stream.
- **Crucial Rule:** Every directory contains special entries for the current directory (`.`) and the parent directory (`..`). You must explicitly check for these names and skip them; otherwise, your program will fall into an infinite recursive loop.

3. Constructing Full Paths and Checking Metadata (`stat`)

- `readdir()` only provides the relative name of the entry (e.g., `file.txt` or `subfolder`), not its full path. You must concatenate the current directory path with the entry name to form a complete path string.
- Pass this full path to `lstat()` or `stat()` to retrieve the file metadata structure.

4. Handling Files vs. Subdirectories

- Examine the file type field inside the metadata structure (or use the helper macros provided by the system).

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4. Handling Files vs. Subdirectories

- Examine the file type field inside the metadata structure (or use the helper macros provided by the system).
- **If it is a regular file:** Perform your desired action (such as printing the name or processing its contents).
- **If it is a subdirectory:** Recursively call your traversal function, passing this new subdirectory path as the argument.

5. Cleanup

- Once the loop finishes reading all entries, close the directory stream using `closedir()` to prevent file descriptor leaks.

Important Best Practices

- **Buffer Safety:** When combining paths, ensure your path buffers are large enough to prevent overflow if the directory tree is deeply nested.
- **Error Handling:** Always check the return values of your system calls, as permission denied errors are very common when scanning arbitrary file system trees.

How do I differentiate between files and directories when searching, using flags like `-f` (file only) and `-d` (directory only)

To differentiate between files and directories when searching using custom command-line flags like `-f` (files only) and `-d` (directories only), your program needs to combine argument parsing with metadata inspection.

Here is the conceptual breakdown of how to implement this logic:

1. Parsing the Flags

- When your program starts, inspect the command-line arguments (`argc` and `argv`) to see if the user specified `-f` or `-d` .
- Set boolean state variables (for example, `search_files_only` or `search_dirs_only`) based on which flag is detected, so the rest of your program knows what constraint to enforce.

2. Inspecting Item Metadata

- As your search or traversal function encounters items, it builds the full path for each entry.
- Pass that full path into a metadata-retrieval function (such as `stat()` or `lstat()`). This populates a system structure containing details about the item, including its permission and type bits.

3. Evaluating File Types

- POSIX provides standard macro functions that evaluate the metadata structure to determine whether an item is a regular file or a directory.
- You evaluate these checks against the item's mode field to classify it cleanly.

4. Applying the Filter Logic

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4. Applying the Filter Logic

Before performing an action on an item (like printing it or recursing into it), apply your search flags:

- **If `-f` is active:** Skip the item if it evaluates to a directory, processing only regular files.
- **If `-d` is active:** Skip regular files, processing only directories (and continuing recursion into them).
- **If no flag is provided:** Process both files and directories normally.

Would you like to know how to use `lstat()` instead of `stat()` to safely handle symbolic links without following them?

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How do I execute an external command like `ls` or `gcc` in C using `fork()` and `execvp()`?

Executing external commands in C requires a cooperative dance between process creation, program replacement, and process synchronization.

Here is the conceptual breakdown of how this works using `fork()` and `execvp()` :

1. Splitting the Process (`fork()`)

To run a new program without terminating your own shell or application, your program must first duplicate itself using the `fork()` system call.

- **The Forking Action:** This splits your program into two identical running copies: the **parent process** (your original program) and the **child process** (the clone).
- **The Return Value:** Both processes continue running from the exact same line of code, but `fork()` returns a different value to each. A return value of zero indicates you are currently inside the child process, while a positive value (the child's process ID) indicates you are in the parent process.

2. Transforming the Child (`execvp()`)

Once inside the child process branch, you prepare the command you want to run and pass it to `execvp()` .

- **What it does:** The `execvp()` function takes the name of the executable (like `ls` or `gcc`) and an array of string arguments (where the first element is the command itself and the last element is a null pointer). It then completely replaces the child process's memory space, code, and data with the new program.
- **No Return on Success:** If `execvp()` succeeds, the original instructions of your child process are wiped out, meaning it **never returns** to the line after the call. If it does return, it means an error occurred (such as the command not being found).

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3. Managing the Parent (`wait()`)

While the child process goes off to execute the external command, the parent process needs to know what is happening.

- **Synchronizing:** The parent process typically calls a waiting function (like `wait()` or `waitpid()`) to pause its own execution until the child process finishes its task.
- **Cleanup:** This ensures the operating system can clean up the child process resources properly and allows the parent to inspect the exit status of the external command once it completes.

Would you like to explore how to redirect the standard input or output streams of the child process (like piping or file redirection)?

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What's the difference between `execvp`, `execv`, `execlp`, and `execve`? Which one should I use?

The `exec` family of functions is used to replace the current process image with a new process image. The suffix letters (`l` , `v` , `p` , `e`) dictate how arguments are passed, whether the system path is searched, and how environment variables are handled.

Understanding the Suffix Letters

- `l` (**List**): Arguments are passed as a variadic list of individual string pointers ending with a null pointer. Use this when you know the exact number of arguments statically at compile time.
- `v` (**Vector**): Arguments are passed as an array of string pointers (a vector). Use this when your arguments are already stored in an array or dynamically generated (like tokenized command strings).
- `p` (**Path**): The function searches the system `PATH` environment variable to locate the executable. This means you only need to provide the command name (such as `ls`), rather than its absolute path (`/bin/ls`).
- `e` (**Environment**): The function allows you to pass a custom array of environment variables explicitly, rather than automatically inheriting the current process's environment.

Breakdown of the Specific Functions

- `execv` : Requires the full or relative path to the executable and an argument array. It does not search the `PATH` and uses the current environment.
- `execvp` : Uses the system `PATH` to find the executable and takes an argument array. It uses the current environment.
- `execlp` : Uses the system `PATH` to find the executable and takes an argument list. It uses the current environment.
- `execve` : The core system call. It requires the full or relative path, an argument array, and a custom environment array. None of the other helpers allow custom environments

Understanding the Unix Letters

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- `execve` : The core system call. It requires the full or relative path, an argument array, and a custom environment array. None of the other helpers allow custom environments directly except this one.

Which One Should You Use?

For building a custom shell or a program that needs to run external commands like a user typed them into a terminal, `execvp` is almost always the best choice.

It handles `PATH` lookup automatically so users do not have to type out full file paths, and it accepts an array of string tokens, which aligns perfectly with how argument lists are structured after tokenizing user input.

1. Bypassing the Wait Function

- **The Parent's Role:** When you fork a process for a foreground command, the parent immediately calls a waiting function and pauses until that child finishes. For a background command, the parent process skips this waiting function entirely.
- **Immediate Resumption:** After calling `fork()`, the parent process immediately loops back to accept the next user input or command, leaving the child process running concurrently in the background.

2. Executing in the Child

- Inside the child process branch, your program replaces its execution context with the target command using an `exec` function.
- Because the parent is no longer synchronized with this child, the operating system manages its execution independently.

3. The Zombie Process Challenge

When a background process finishes executing, it terminates, but its exit status remains stored in the operating system's process table. If a parent process never calls a waiting function to collect this status, the finished child becomes a **zombie process**, consuming system resources until your program eventually exits.

To handle this cleanly without blocking your shell loop, programs typically use one of two strategies:

- **Signal Handling:** Setting up a handler for the signal that the operating system sends when a child process terminates, instructing the system to automatically clean up finished background tasks.
- **Ignoring the Signal:** Explicitly telling the operating system to discard the termination status of child processes automatically, preventing zombies from accumulating altogether.

How do `dup`, `dup2`, and file descriptors work in C? Explain with a simple example.

What is a File Descriptor?

A **file descriptor** is simply a small, non-negative integer that the operating system uses to track and manage open files, network sockets, or input/output streams.

When any program runs, the system automatically opens three standard descriptors for it:

- **0: Standard Input** (`stdin`)
- **1: Standard Output** (`stdout`)
- **2: Standard Error** (`stderr`)

How `dup()` Works

The `dup()` function takes an existing, open file descriptor and creates a **copy** of it.

- **Shared Access:** Both the original and the new file descriptor point to the exact same underlying open file description. They share the same file position pointer and access modes.
- **Lowest Available Number:** The operating system automatically assigns the *lowest numbered* unused integer available in your process's file descriptor table for the new copy.

How `dup2()` Works

The `dup2()` function also duplicates a file descriptor, but it gives you **explicit control** over what number the new descriptor will be.

- **Target Control:** You pass both the existing file descriptor and a desired target integer to

How `dup2()` Works

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- **Target Control:** You pass both the existing file descriptor and a desired target integer to the function.
- **Automatic Overwriting:** If the target integer is already open (for instance, if you target standard output, which is normally 1), the system automatically closes that target file descriptor first, and then safely replaces it with a clone of your source descriptor.

Conceptual Example: Redirecting Output

Imagine you want a program to stop printing text to the terminal screen and instead save everything directly into a text file.

1. **Opening the File:** Your program opens a text file for writing. The operating system hands back a file descriptor integer to represent that file, for example, number `5`.
2. **Duplicating with `dup2`:** You call `dup2`, passing your file descriptor (`5`) and standard output (`1`) as the target.
3. **The Result:** The system closes standard output and makes number `1` a precise clone of file descriptor `5`.
4. **Data Flow:** Now, whenever your program prints text normally—which always goes to standard output (`1`)—that text automatically redirects straight into your text file because standard output has been re-routed.

Implement input redirection (<), output redirection (>), and append redirection (>>) for my shell's commands.

To implement input redirection (<), output redirection (>), and append redirection (>>) in your C shell, you need to manage file opening, handle file descriptors, and manipulate process streams *after* forking the child process, but *before* executing the command with an `exec` function.

Here is the conceptual breakdown of how to implement redirection:

1. Scanning and Stripping Redirection Tokens

When your shell parses the command line, it needs to scan the token list for redirection operators (< , > , >>) followed by a filename token.

- **Extraction:** When your parser encounters one of these symbols, it records the operator type and grabs the very next token as the target filename.
- **Cleanup:** You must remove the redirection operator and its filename from the main argument token array that you pass to your `exec` function. If you leave them in the argument array, the external command (like `ls`) will try to treat the filename as a positional argument and throw an error.

2. Performing Redirection in the Child Process

Redirection must happen inside the **child process** right after `fork()`. If you modified file descriptors in the parent shell process, your shell would lose its own standard input/output.

Depending on which operator was found, the child process opens the target file using standard file system functions:

- **For Input (<):** Open the target file with read-only permissions. If the file doesn't exist, handle the error gracefully.
- **For Output (>):** Open the target file with write-only permissions, creating it if it doesn't exist, and truncating (emptying) it if it already exists.
- **For Append (>>):** Open the target file with write-only permissions, creating it if it

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- **For Append (>>)**: Open the target file with write-only permissions, creating it if it doesn't exist, but setting it to append mode so existing data is preserved.

3. Wiring the Streams with `dup2()`

Once the file is successfully opened, your code uses `dup2()` to wire the file descriptor to the correct standard stream:

- **For < :** Duplicate the file descriptor over standard input (descriptor `0`).
- **For > and >> :** Duplicate the file descriptor over standard output (descriptor `1`).

4. Closing Extra File Descriptors

After calling `dup2()`, the original file descriptor returned by the `open` function is no longer needed because the standard stream points directly to the file. You should close that original file descriptor immediately to avoid leaking file descriptors in the child process.

5. Executing the Cleaned Command

With the standard input or output now silently re-routed to your target file, call your `exec` function with the modified argument list. Any data the program prints or reads will automatically flow through the file instead of the terminal.

How do I implement a two-command pipe like `ls | grep .c` using `pipe()`, `fork()`, and `dup2()` in C?

To implement a two-command pipe like `ls | grep .c` in C, your program needs to orchestrate a data bridge between two separate child processes using the `pipe()` system call along with `dup2()` for redirection.

Here is the conceptual step-by-step breakdown:

1. Creating the Pipe (`pipe()`)

Before creating any child processes, your shell must call the `pipe()` function.

- This system call takes an array of two integers and populates them with two new file descriptors: one designated for reading data, and one designated for writing data.
- Anything written into the write end of the pipe can be read immediately from the read end, creating an in-memory communication channel.

2. Forking the Processes

To run both commands concurrently, your shell needs to create two separate child processes—one for the left-hand command (`ls`) and one for the right-hand command (`grep .c`). You will typically call `fork()` twice, or fork twice sequentially, to spawn both workers.

3. Configuring the First Child (The Writer)

In the child process responsible for the first command (`ls`):

- You want its standard output to flow into the pipe instead of printing to the terminal screen.
- You use `dup2()` to duplicate the pipe's **write end** over standard output (file descriptor 1).
- You close both of the original pipe file descriptors in this child process so they don't linger or interfere.
- Finally, you execute the first command using an `exec` function.

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In the child process responsible for the first command (`ls`):

- You want its standard output to flow into the pipe instead of printing to the terminal screen.
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- You close both of the original pipe file descriptors in this child process so they don't linger or interfere.
- Finally, you execute the first command using an `exec` function.

4. Configuring the Second Child (The Reader)

In the child process responsible for the second command (`grep .c`):

- You want its standard input to pull from the pipe instead of waiting for keyboard input.
- You use `dup2()` to duplicate the pipe's **read end** over standard input (file descriptor 0).
- You close both of the original pipe file descriptors in this child process.
- Finally, you execute the second command using an `exec` function.

5. Parent Process Cleanup and Waiting

The parent shell process does not read from or write to the pipe itself, which is a critical detail:

- **Closing Descriptors:** The parent **must** close both the read and write ends of the pipe immediately after forking the children. If the parent leaves the write end open, the second command (`grep`) will hang indefinitely, waiting for an end-of-file marker that will never arrive because an open reference still exists.
- **Synchronizing:** The parent calls a waiting function twice (once for each child process) to ensure both commands finish executing completely before the shell prints its next prompt.

how to check if the current directory path points to something

Worked for 32s >

To check if the current working directory path is valid in C, you can check the return value of `getcwd()` (from `<unistd.h>`) or test the path using `access()` / `stat()`.

1. Checking the return value of `getcwd()`

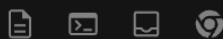
`getcwd(buf, size)` returns a pointer to `buf` if successful, or `NULL` if it fails (e.g., if the current directory was deleted or permissions were lost):

```
c
char cwd[1024];

if (getcwd(cwd, sizeof(cwd)) != NULL) {
    // Successfully retrieved path! `cwd` points to a valid
    // directory string.
    printf("Current working directory: %s\n", cwd);
} else {
    // getcwd failed (e.g. current directory was deleted)
    perror("getcwd() error");
}
```

2. Checking if any path exists on disk using `access()` or `stat()`

If you have a path string and want to check if it points to an existing file or directory on the filesystem:



Review Changes

of the *previous* pipe (`pipes[i - 1][0]`), and redirect standard output to the write end of the *current* pipe (`pipes[i][1]`).

- For the last command ($i = N - 1$): Redirect standard input to the read end of the *previous* pipe (`pipes[i - 1][0]`), and leave standard output alone (or handle output redirection `> / >>`).
5. **Parent Cleanup:** The parent process must close *all* pipe file descriptors after spawning the children, then loop `wait()` to reap every child process.

Code Example

C



```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

// Example function skeleton to execute N commands connected by pipes
void execute_pipeline(char ***cmd_args, int num_cmds) {
    int num_pipes = num_cmds - 1;
    int pipes[num_pipes][2];

    // 1. Create all necessary pipes
    for (int i = 0; i < num_pipes; i++) {
        if (pipe(pipes[i]) < 0) {
            perror("pipe failed");
            exit(1);
        }
    }

    // 2. Fork a process for each command
    for (int i = 0; i < num_cmds; i++) {
        pid_t pid = fork();
        if (pid < 0) {
            perror("fork failed");
            exit(1);
        }
    }
}
```

```

    if (pid < 0) {
        perror("fork failed");
        exit(1);
    }

    if (pid == 0) {
        // --- CHILD PROCESS ---

        // If not the first command, redirect stdin from the previous pipe's read end
        if (i > 0) {
            dup2(pipes[i - 1][0], STDIN_FILENO);
        }

        // If not the last command, redirect stdout to the current pipe's write end
        if (i < num_pipes) {
            dup2(pipes[i][1], STDOUT_FILENO);
        }

        // Close all pipe copies in the child process
        for (int j = 0; j < num_pipes; j++) {
            close(pipes[j][0]);
            close(pipes[j][1]);
        }

        // Execute the command
        execvp(cmd_args[i][0], cmd_args[i]);
        perror("execvp failed");
        exit(1);
    }
}

// --- PARENT PROCESS ---

// Parent closes all pipe ends so children don't hang waiting for EOF
for (int i = 0; i < num_pipes; i++) {
    close(pipes[i][0]);
    close(pipes[i][1]);
}

// Wait for all child processes to finish

```

```

    }

    // If not the last command, redirect stdout to the current pipe's write end
    if (i < num_pipes) {
        dup2(pipes[i][1], STDOUT_FILENO);
    }

    // Close all pipe copies in the child process
    for (int j = 0; j < num_pipes; j++) {
        close(pipes[j][0]);
        close(pipes[j][1]);
    }

    // Execute the command
    execvp(cmd_args[i][0], cmd_args[i]);
    perror("execvp failed");
    exit(1);
}

// --- PARENT PROCESS ---

// Parent closes all pipe ends so children don't hang waiting for EOF
for (int i = 0; i < num_pipes; i++) {
    close(pipes[i][0]);
    close(pipes[i][1]);
}

// Wait for all child processes to finish
for (int i = 0; i < num_cmds; i++) {
    wait(NULL);
}
}

```

Would you like to discuss how to parse a command string into that 2D command-argument array (`cmd_args`) using the pipe character as a delimiter?

How do I handle `;` to run multiple commands sequentially in my shell — e.g., `echo hi ; ls ; pwd`?

To handle command separators like `;` to run multiple commands sequentially (such as `echo hi ; ls ; pwd`), your shell needs to parse the token list into separate command blocks and execute them one after another, waiting for each command to finish before starting the next.

Here is the conceptual breakdown of how sequential execution works:

1. Splitting by the Semicolon Operator

When your shell parses the command line, it scans your tokenized array for the semicolon (`;`) symbol.

- **Command Grouping:** Instead of treating the entire line as a single massive command, the semicolon acts as a boundary.
- **Sub-arrays:** You split your full list of tokens into distinct, smaller sub-arrays of arguments —each representing an independent command block ready for execution.

2. Sequential Execution Loop

Once you have your separated list of command blocks, your shell enters a simple loop:

- **Fork and Execute:** Take the first command block, call `fork()` to create a child process, and use an `exec` function to run it.
- **Synchronous Waiting:** Unlike background execution where the parent immediately moves on, the parent process for a sequential command **must call a waiting function** (like `waitpid()`) and block until that specific child process completely finishes.
- **Proceeding to the Next:** Only after the current command has finished executing does the loop move on to the next command block in the sequence, repeating the exact same fork-and-wait cycle.

how to check if the current directory path points to something

Worked for 32s >

To check if the current working directory path is valid in C, you can check the return value of `getcwd()` (from `<unistd.h>`) or test the path using `access()` / `stat()`.

1. Checking the return value of `getcwd()`

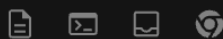
`getcwd(buf, size)` returns a pointer to `buf` if successful, or `NULL` if it fails (e.g., if the current directory was deleted or permissions were lost):

```
c
char cwd[1024];

if (getcwd(cwd, sizeof(cwd)) != NULL) {
    // Successfully retrieved path! `cwd` points to a valid
    // directory string.
    printf("Current working directory: %s\n", cwd);
} else {
    // getcwd failed (e.g. current directory was deleted)
    perror("getcwd() error");
}
```

2. Checking if any path exists on disk using `access()` or `stat()`

If you have a path string and want to check if it points to an existing file or directory on the filesystem:



Review Changes

2. Checking if any path exists on disk using `access()` or `stat()`

If you have a path string and want to check if it points to an existing file or directory on the filesystem:

```
c
#include <unistd.h>

if (access(cwd, F_OK) == 0) {
    // The path exists!
} else {
    // The path does not exist or is inaccessible
}
```

To verify specifically that it is a **directory**:

```
c
#include <sys/stat.h>

struct stat st;
if (stat(cwd, &st) == 0 && S_ISDIR(st.st_mode)) {
    // Path exists AND is a directory
}
```

Fixed `display_prompt()` in `c` `src/prompt.c`

how to check if the current directory is path to osn mp 1

Worked for 19s >

According to the **OSN Mini Project 1 (C-Shell) specification**, the prompt home directory (`~`) is defined as the directory from which the shell executable was launched.

Here is how to check if the current working directory (`full_path`) is inside or equal to the shell's root directory (`shell_home_dir`):

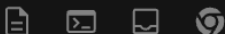
Logic to Check the Path

```
c
size_t home_len = strlen(shell_home_dir);

// 1. Exact match -> Current directory IS the shell root (~):
if (strcmp(full_path, shell_home_dir) == 0) {
    snprintf(out_buf, buf_size, "~");
}

// 2. Subdirectory -> Inside shell root (~/subdir):
else if (home_len > 0 && strncmp(full_path, shell_home_dir,
home_len) == 0 && full_path[home_len] == '/') {
    snprintf(out_buf, buf_size, "~%s", full_path + home_len);
}

// 3. Outside shell root -> Full absolute path:
else {
    strncpy(out_buf, full_path, buf_size - 1);
    out_buf[buf_size - 1] = '\0';
}
```



Review Changes